



School of Computer Science Engineering and Information Systems

UCSC318L: Exploratory Data Analytics

WINSEM 2025-2026

A Report on the Course Project

Project Title:

- SMART ANALYSIS AND PREDICTION OF LUNG, SKIN, AND KIDNEY DISEASES USING DATA TRANSFORMATION, EXPLORATORY DATA ANALYTICS, AND MACHINE LEARNING

Team Member(s) with Reg No and Name:

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SLOT: E1 + TE1

1. Problem Statement

In recent years, health-related issues such as lung disease, skin disorders, and kidney diseases have increased significantly due to environmental pollution, unhealthy lifestyle habits, and genetic factors. Early detection and analysis of these diseases are essential to reduce risks and improve treatment outcomes.

This project aims to perform Exploratory Data Analytics (EDA) on medical datasets to identify patterns and relationships between patient health conditions and disease occurrence. The study focuses on analyzing factors such as age, smoking habits, alcohol consumption, blood pressure, sugar levels, and cholesterol.

The objectives of this project include:

- * Understanding disease patterns
- * Identifying key risk factors
- * Analyzing relationships between health indicators
- * Building predictive models using machine learning

This analysis helps in supporting early diagnosis and improving healthcare decision-making.

2. DATA SET

Dataset 1: Lung Disease Dataset

This dataset contains patient information related to lung health.

Key Features:

- * Patient ID
- * Age
- * Gender
- * Smoking Habit
- * Body Mass Index (BMI)
- * Lung Score
- * Disease Status (Target Variable)

Dataset 2: Kidney and Skin Disease Dataset

This dataset provides additional health parameters.

Key Features:

- * Patient ID
- * Blood Pressure
- * Sugar Level
- * Cholesterol Level
- * Alcohol Consumption
- * Skin Condition

Dataset Description:

- * Both datasets are structured and contain numerical and categorical data
- * The datasets were merged using *patient_id*
- * Final dataset contains combined health information for analysis

Dataset Challenges:

- * Missing values
- * Categorical variables
- * Data integration complexity
- * Slight imbalance in disease distribution

3. DATA TRANSFORMATION TECHNIQUES USED :

Data transformation plays a crucial role in preparing data for analysis and machine learning.

Steps Performed:

1. Data Cleaning

- * Handled missing values using median imputation
- * Replaced invalid values
- * Ensured consistency

2. Dataset Merging

- * Combined lung dataset and kidney-skin dataset
- * Used *patient_id* as the key
- * Created a unified dataset

3. Encoding of Categorical Variables

Categorical features such as:

- * Gender

- * Smoking
- * Alcohol
- * Skin Condition

4. Feature Scaling

- * Applied *StandardScaler*
- * Normalized all numerical features

5. Feature Selection

- * Selected important attributes affecting disease prediction
- * Removed unnecessary data

Importance of Data Transformation:

- * Improves data quality
- * Enhances model performance
- * Reduces noise
- * Enables accurate predictions

4. VISUAL AIDS (PLOTS) :

Various visualizations were used to understand the dataset:

Histogram

- * Used to analyze age, BMI, and blood pressure distribution
- * Helps identify patterns and spread

Bar Chart

- * Shows disease vs healthy distribution
- * Helps understand class imbalance

Count Plot

- * Smoking vs Disease
- * Alcohol vs Disease
- * Shows impact of lifestyle

Box Plot

- * Sugar level vs disease
- * Cholesterol vs disease
- * Identifies outliers and variation

Scatter Plot

- * Age vs Disease
- * Shows relationship between variables

Correlation Heatmap

- * Displays relationships between features
- * Helps identify important variables

Pairplot

- * Shows multiple feature interactions
- * Helps in deeper analysis

Major Uses of Plots:

- * Understanding data distribution
- * Identifying relationships
- * Detecting outliers
- * Finding patterns

Key Insights from Visualizations:

- * Smoking increases risk of lung disease
- * High sugar levels are linked to kidney disease
- * High BMI is associated with health risks
- * Age plays an important role in disease occurrence

5. MACHINE LEARNING MODEL:

A. Models Implemented

1. Logistic Regression

- * Simple and interpretable
- * Suitable for classification

2. Decision Tree

- * Handles complex relationships
- * Easy to visualize

3. Random Forest

- * Ensemble model
- * Provides high accuracy
- * Reduces overfitting

B. Model Workflow

1. Data preprocessing
2. Dataset merging
3. Feature scaling
4. Train-test split
5. Model training
6. Prediction
7. Evaluation

C. Working Mechanism

- * Input features are given to models
- * Models learn patterns from training data
- * Predictions are made on test data
- * Output is compared with actual values

6. PERFORMANCE MEASURES :

Evaluation Metrics Used:

- * Accuracy
- * Precision
- * Recall
- * F1-Score

Confusion Matrix

Used to analyze:

- * True Positives
- * True Negatives
- * False Positives
- * False Negatives

Model Comparison:

Model	Performance
Logistic Regression	Good
Decision Tree	Moderate
Random Forest	Best

Model Reliability

- * Random Forest performed best due to handling complex data
- * More reliable for prediction

7. CONCLUSION:

This project successfully applied Exploratory Data Analytics and machine learning techniques to analyze lung, skin, and kidney diseases.

The results show that:

- * Lifestyle factors such as smoking and alcohol significantly affect health
- * Medical parameters like sugar level, BMI, and blood pressure influence diseases
- * Machine learning models can effectively predict disease occurrence

Among all models, *Random Forest achieved the best performance*.

Overall, this project provides valuable insights for early disease detection and supports better healthcare decisions. Future improvements can include real-world datasets and advanced deep learning models.

Logistic Regression Accuracy: **0.67**

Decision Tree Accuracy: **0.58**

Random Forest Accuracy: **0.65**

Logistic Regression Performance:

Accuracy: **0.67**

Precision: **0.4489**

Recall: **0.67**

F1 Score: **0.5376047904191616**

Decision Tree Performance:

Accuracy: **0.58**

Precision: **0.5597260273972603**

Recall: **0.58**

F1 Score: **0.568**

Random Forest Performance:

Accuracy: **0.65**

Precision: **0.5590425531914893**

Recall: **0.65**

F1 Score: **0.5581939799331104**

Model Accuracy

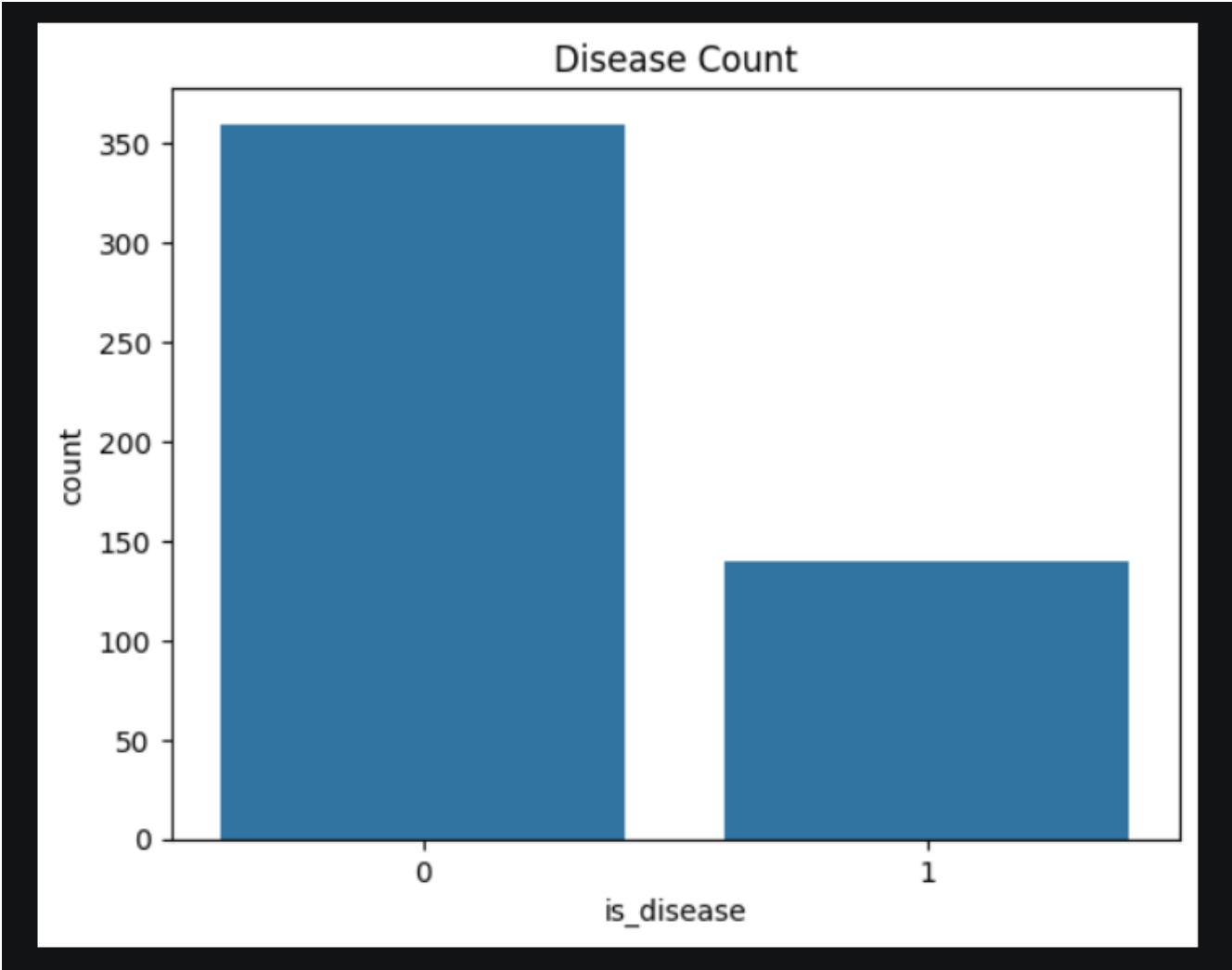
0 Logistic Regression 0.67

1 Decision Tree 0.58

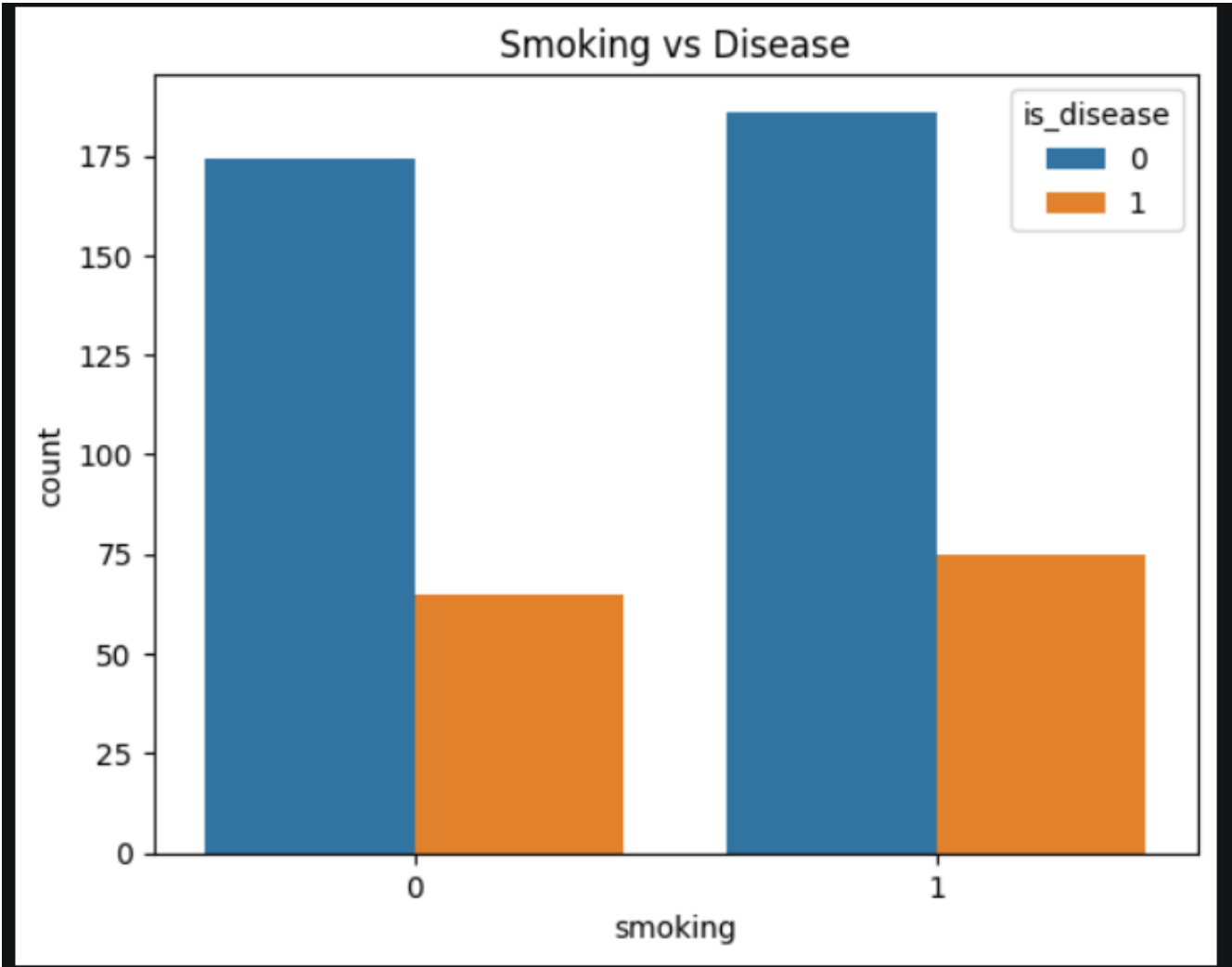
2 Random Forest 0.65

Plots:

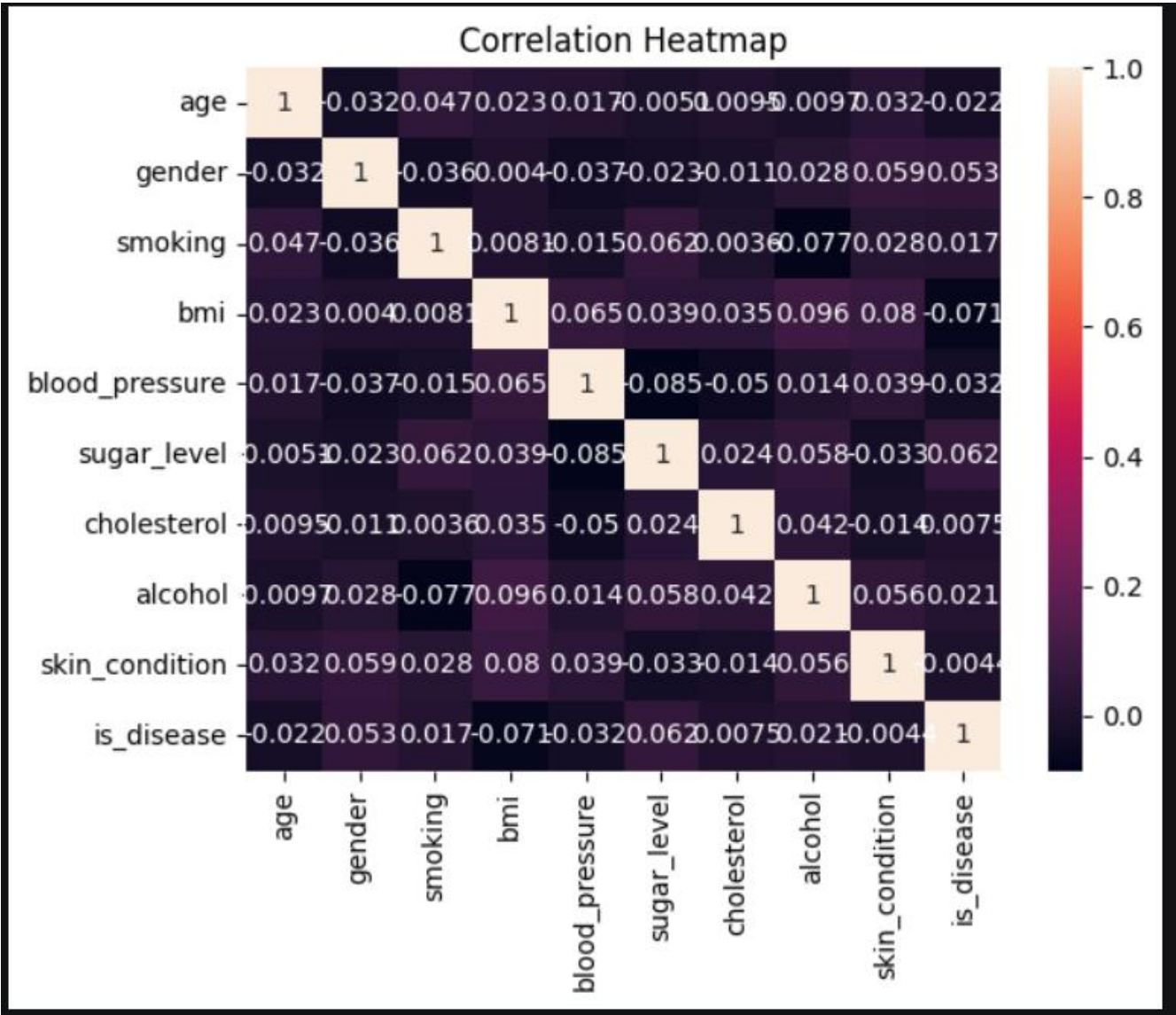
Count Plot:



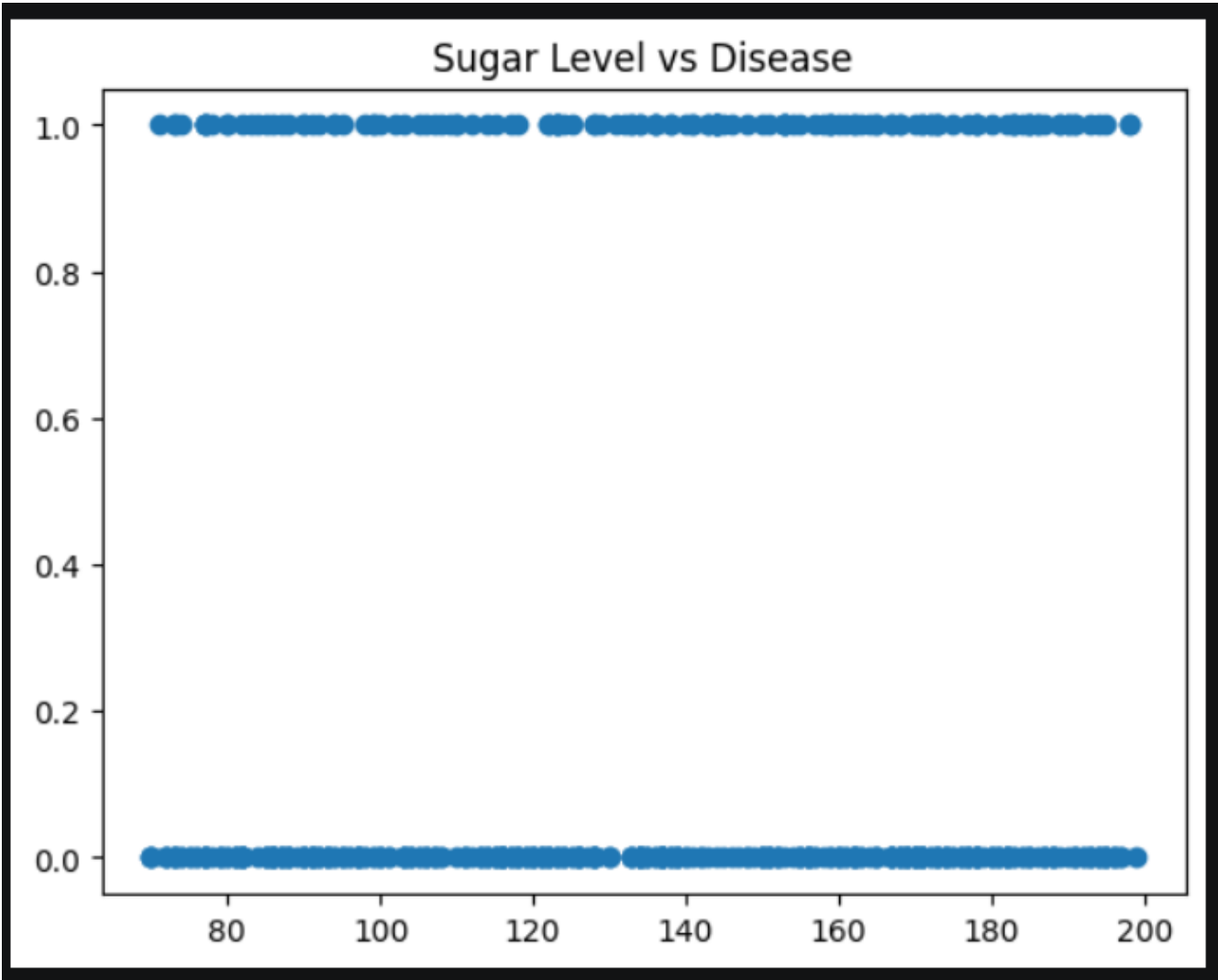
Bar Chart:



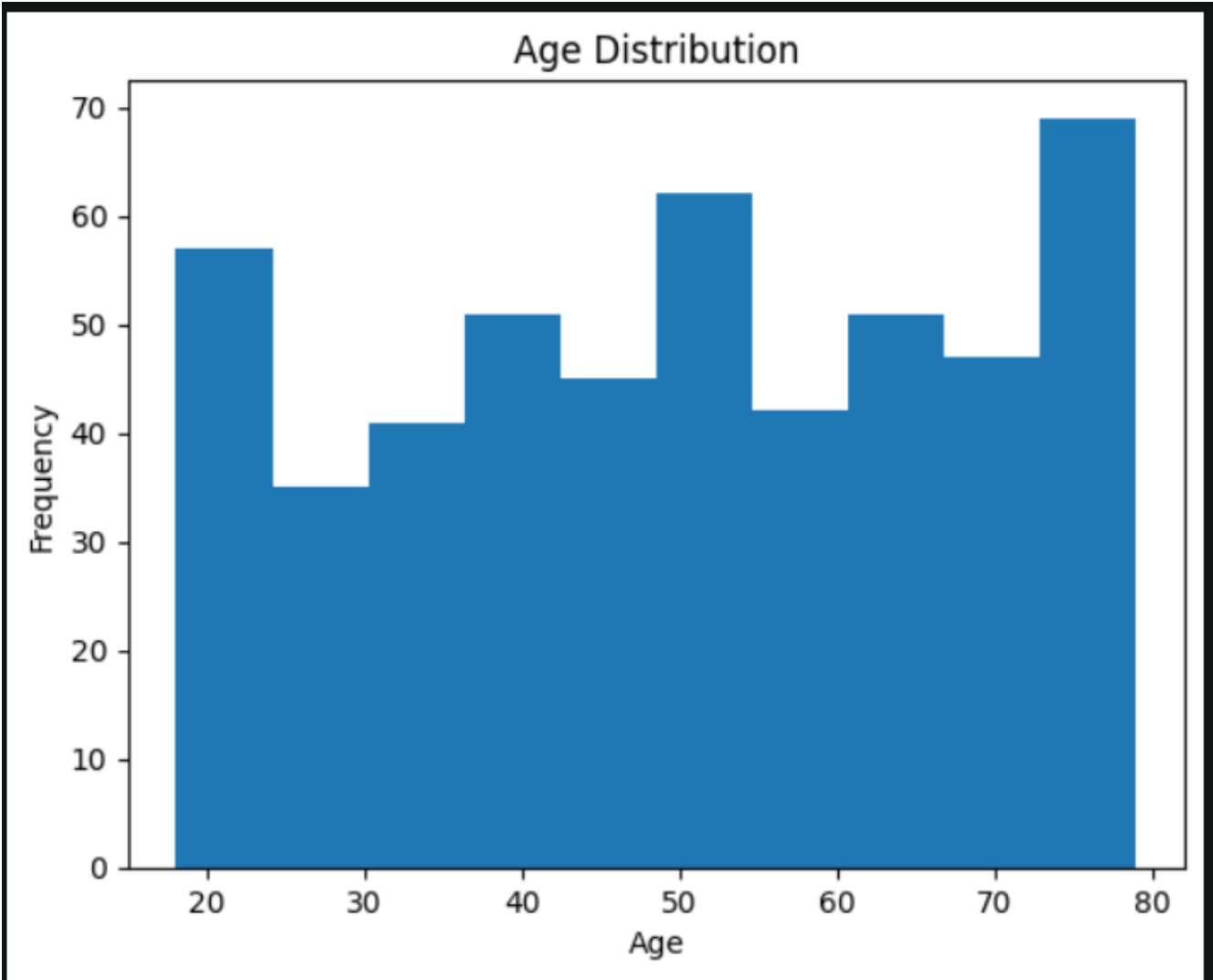
Heat Map:



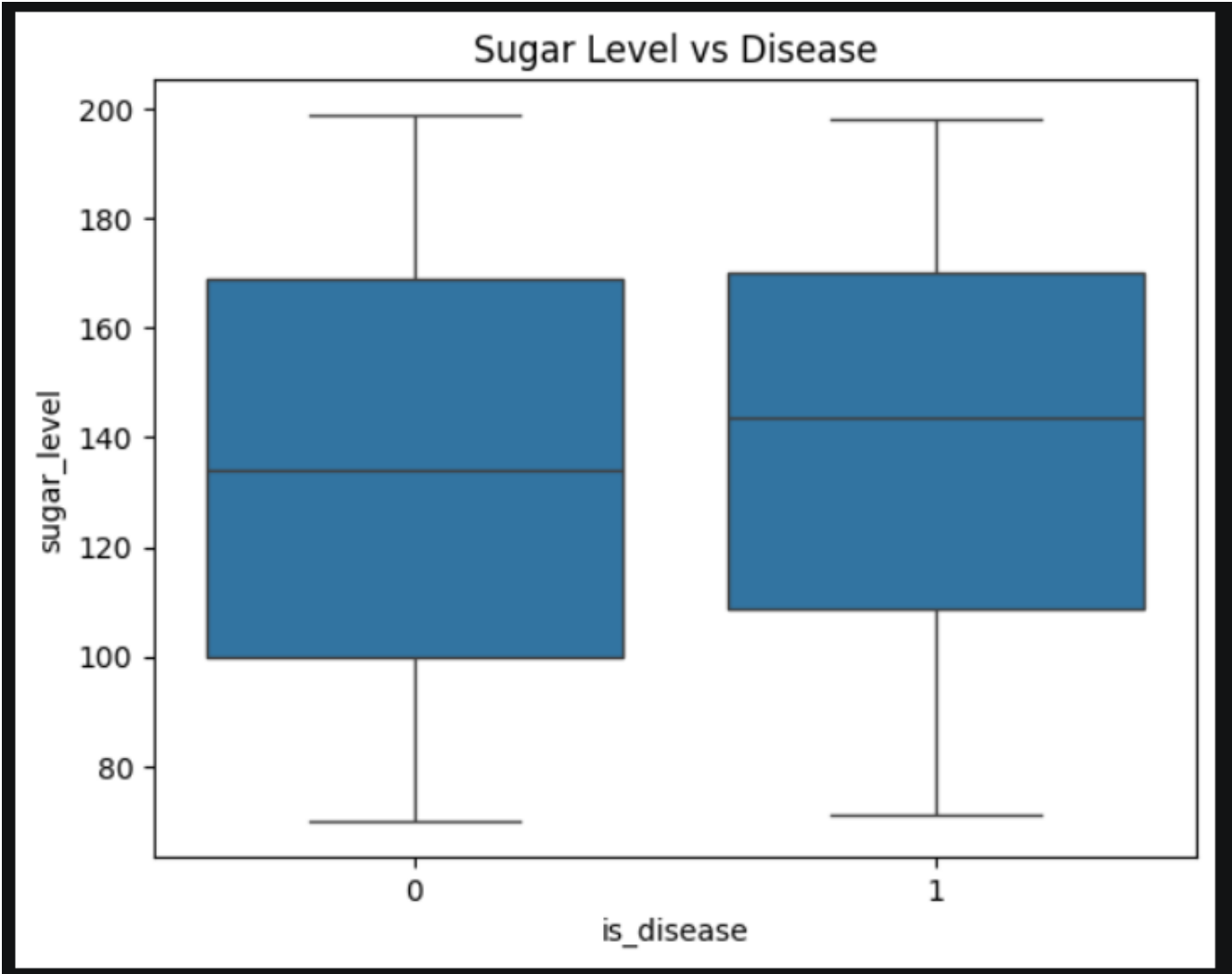
Scatter Plot:



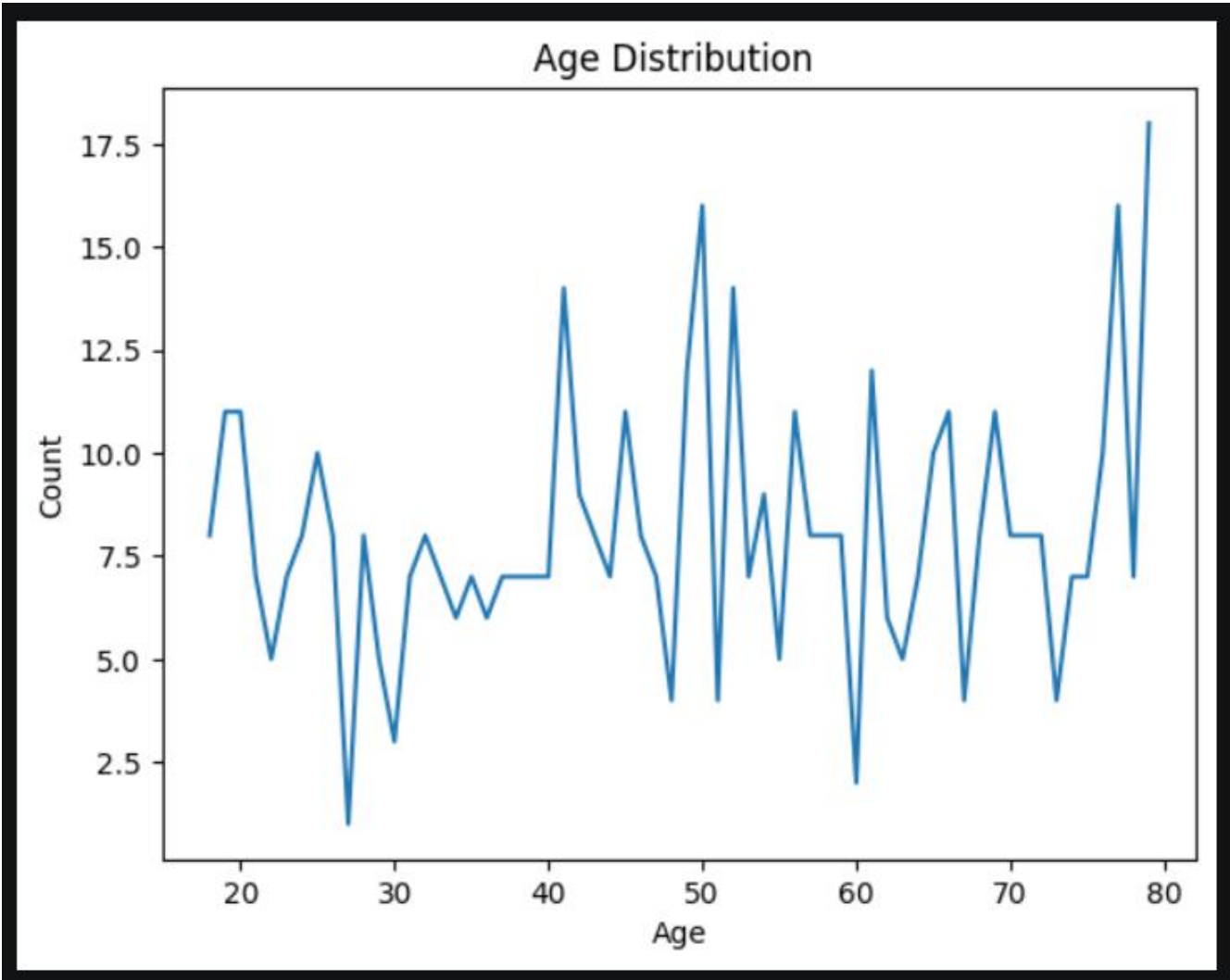
Histogram:



Box Plot:



Line Chart:



Pie Chart:

